

EXECUTIVE SYSTEM ARCHITECTURE & MASTER SPECIFICATION PROMPT

MAMAL PLATFORM

The Higgsfield.ai of Marketing & Commerce — Unified Generative Multimodal Execution with Predictive Human Behavior Intelligence

Target Directory:
`/platform`

Lineage Consolidation: 22 Source
Repositories → 6 Unified Tools

Architecture: Edge-Augmented
Hybrid Monorepo

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AUTONOMOUS EXECUTION DIRECTIVE

You are an elite Principal Systems Architect, AI/ML Research Engineer, and Product Design Director. You are tasked with engineering `/platform`. This document serves as the exhaustive, definitive master prompt and architectural specification. Every tool, predictive model pipeline, closed-loop automation, and UI token defined herein must be built with uncompromising precision.

1. SYSTEM IDENTITY & CORE MISSION

Mamal Platform is the industry's first unified platform merging **generative multimodal marketing** with **real-time predictive human behavior analytics for commerce**.

Traditional SaaS landscapes force merchants and brands to license disparate, uncoordinated point solutions: one tool for SEO audits, another for social proof, separate apps for short links and file sharing, external agencies for ad creative generation, third-party services for uptime monitoring, and complex analytics suites for traffic recording. None of these systems communicate with each other.

The Higgsfield.ai of Commerce closes this gap. It creates a closed-loop system where:

- **Predictive Intelligence:** Continuously measures and forecasts human intent (purchase propensity, churn risk, price sensitivity, hook retention, and brand sentiment).
- **Generative Actuation:** Automatically synthesizes and executes marketing countermeasures (Seedance 2.5 video ads, Nanobanana product photography, ElevenLabs audio dubbing, automated trending articles, dynamic social proof popups, and smart routing short links).
- **Unified Foundation:** Unifies 22 legacy SaaS products into 6 core tools sharing a single workspace, unified billing, universal credit currency, and an asynchronous event bus (`@mamal/bus`).

2. PREDICTIVE ANALYTICS ENGINE FOR HUMAN BEHAVIOR

The predictive engine shifts businesses from *reactive historical reporting* to *prescriptive pre-emption*. It feeds clickstream telemetry, session recordings, checkout friction, and social signals into specialized machine learning models to forecast future outcomes.

2.1 The Eight Core Modeled Human Behaviors

Behavioral Dimension	Telemetry & Behavioral Feature Signals	Predictive Output & Actuation
1. High-Propensity Checkout Intent P(BUY) : 0.00 – 1.00	Cursor velocity deceleration over purchase triggers; tab-switching frequency across tier cards; scroll velocity on guarantee blocks; cart-mutation cadence.	Triggers dynamic friction-removal interventions, localized trust badges, or instant checkout guarantees before session termination.
2. Price Sensitivity & WTP Elasticity QUOTIENT R_PRICE	Dwell-time comparison across pricing plans; coupon code trial cadence; quantity selector toggling; inbound acquisition channel affinity.	Calculates maximum non-discounted willingness-to-pay (WTP). Issues volume tiers or dynamic incentives without sacrificing gross margins.
3. Churn & Disengagement Trajectory P(CHURN) HAZARD RATE	Inter-session latency lengthening (>2.5× historical baseline); feature breadth decay; falling email/push open rates; unvoiced negative feedback.	7, 30, and 90-day churn probability curves. Triggers proactive retention concierge workflows and automated feedback recovery forms.
4. Predictive Lifetime Value 365-DAY PLTV FORECAST	First-action completion velocity; initial basket gross margin; catalog exploration diversity; referral source baseline quality.	Segments incoming visitors into VIP, Standard, or Low-Yield cohorts. Optimizes ad bidding and dedicated support routing.
5. Brand & Persona Receptivity AFFINITY VECTOR	Engagement latency on founder-led video vs. catalog stills; sentiment tone on comments; bio-link avatar click paths; UGC interaction depth.	Determines whether downstream marketing delivers humanized storytelling (personal brand) or corporate enterprise feature breakdowns.
6. Creative Scroll-Stop & Hook Retention HOOK INDEX H_EFF	First 3-second video playback completion; viewport scroll acceleration across social feeds; micro-scrubbing gestures; sound-on toggle events.	Forecasts creative hook capture probability across TikTok, Instagram Reels, and YouTube Shorts before deploying ad budgets.
7. Viral Resonance & UGC Adoption K-FACTOR PREDICTION	Social re-share velocity; bio-link bookmark frequency; coupon claim pass-through rate; quote inquiries generated from shared links.	Predicts viral coefficient. Recommends when to inject paid ad spend behind high-momentum organic social posts.
8. Ad Creative Fatigue Velocity FATIGUE INDEX & DECAY RATE	Frequency-to-CTR degradation slope; negative ad feedback on Meta/TikTok placements; rolling 48-hour CTR variance.	Forecasts remaining creative half-life (hours/days) before ROAS falls below target threshold. Automates variant swapping.

2.2 End-to-End Machine Learning Pipeline

- Data Collection:** Raw event streams from `/track` (cookieless telemetry, session replays, DOM heatmaps), `/confirm` (widget impressions, form submissions, coupon claims), `/market` (ad metrics, watch time), `/link` (short URL hops, QR scans), `/monitor` (latency impacts), and `/audit` (SEO health).
- Data Preparation:** Edge sanitization stripping PII; session stitching resolving anonymous sessions to authenticated customer profiles; temporal aggregation across 1m, 1h, and 24h windows; Z-score feature normalization.
- Feature Engineering:** Recency-Frequency-Monetary with Engagement Depth (RFM+); Gaussian-smoothed attention heatmaps; session navigation DAGs (Directed Acyclic Graphs).
- Model Training & Inference:**
 - Behavioral Forecasting:** Default model: **OpenAI Astra** for high-dimensional consumer behavioral simulation.
 - Causal Reasoning & Attribution:** Default model: **DeepSeek Pro / R1** for complex root-cause churn analysis and multi-touch revenue attribution.
 - Edge Propensity Scoring:** Lightweight ONNX/XGBoost classifiers compiled to WebAssembly running on Cloudflare Workers edge nodes (<15ms latency).

5. **Evaluation & Calibration:** Monitored continuously via Brier scores and AUC-ROC curves. If trailing 7-day conversion prediction accuracy drops below 92%, automated fine-tuning jobs are dispatched via BullMQ.

3. CLOSED-LOOP PREDICTIVE AUTOMATION ENGINE

Predictive scores drive direct, automated business actions across five key operational vectors, eliminating latency between insight and execution:

Revenue Optimization Automations

- **Exit-Intent Interception:** When `p(buy) > 0.60` and cursor velocity indicates impending exit, automatically render a localized `/confirm` modal with a decaying 10% coupon code and live social proof.
- **Predictive Cart Recovery:** If an abandoned checkout has estimated `pLTV > $500`, dispatch a multi-step Web Push drip campaign via `/confirm` at the customer's predicted peak activity hour.
- **Loyalty Re-engagement:** When in-store QR stamp count is within 1 stamp of reward unlock and visit frequency slips, issue automated double-stamp passes valid for 48 hours.

Customer Service & Retention Automations

- **Proactive Churn Neutralization:** If a subscriber's session frequency declines by >50% over 14 days, auto-deliver a personalized `/confirm` private feedback form with high-priority support routing.
- **Negative Review Deflection:** If incoming feedback is ≤ 2 stars, intercept the submission from posting publicly to Google/Facebook, route into a private recovery ticket, alert staff via Slack, and auto-issue a voucher.
- **SLA Proactive Apology:** If `/monitor` detects an outage affecting a client's API, auto-dispatch a transparent status notification and service credit to impacted users.

Article & Editorial Automations (AI SEO & LLMs)

- **Trend-to-Article Auto-Pilot:** Ingest trending keywords from `/market` (trends-checker + DataForSEO). When a query surges >40% WoW, auto-trigger the AI Article Wizard.
- **Autonomous Multi-CMS Publishing:** Synthesizes a 2,500-word, cited, humanized article with contextual Nanobanana diagrams and JSON-LD schema, publishing directly to Shopify, WooCommerce, WordPress, or Ghost.
- **LLM Citation Tracking:** Audit brand citations across ChatGPT, Perplexity, Claude, and Gemini via `/audit`. When competitors gain visibility, auto-update articles to optimize semantic entity grounding.

Advertising & Media Buying Automations

- **Autonomous Creative Refresh:** When ad CTR falls $\geq 25\%$ below its 7-day average, trigger MagicAds to generate 5 new ad copy variations and 3 new Seedance 2.5 video hooks.
- **Multi-Armed Bandit Reallocation:** Dynamically shift ad budgets across Meta, Google, and TikTok based on 3-hour micro-signals, maximizing return on ad spend (ROAS).
- **Competitor Ad Counter-Moves:** Detect competitor pricing shifts via market scanners and instantly draft counter-promotional ad creatives.

3.1 Automation UX: Templates & Visual Node Canvas

- **1-to-3 Click Pre-Built Recipes:** "Turn High-Intent Abandoned Carts into Verified Reviews", "Auto-Draft Weekly Trending Blog & Syndicate to Social Channels", "Intercept Negative Feedback Before Google Reviews", "Auto-Generate Video Ads When Competitor Prices Rise".
- **Drag-and-Drop Visual Automation Canvas:** Interactive DAG graph canvas built with React Flow. Users drag Trigger nodes (events), Predictive Filter nodes (thresholds on `p(buy)`, `p(churn)`), Condition nodes (geo, device), and Action nodes (cross-tool API calls). Features live execution highlights and single-node step testing.

4. DEEP BREAKDOWN OF THE SIX UNIFIED TOOLS

Tool 1: [/audit](#) — Search & AI Visibility Auditor

Consolidates: 66audit, phprank, crawlseo-main, open-seo-main, linkinator-main.

- **72-Rule Technical SEO Engine:** Complete evaluation covering crawlability, semantic HTML, OpenGraph, Core Web Vitals, security headers (HSTS, CSP), and mobile viewport compliance.
- **Bounded-Slice Distributed Crawler:** BullMQ queue workers crawling in slices of ≤ 25 pages. Persists findings atomically. Eliminates crawl loss on worker crashes and streams live page progress to the UI.
- **AI Visibility & LLM Citation Auditing:** Audits brand presence in ChatGPT, Perplexity, Gemini, and Claude search answers. Scores AI share of voice and provides semantic grounding recommendations.
- **18 Free Client-Side Tools:** Unlimited zero-cost tools (robots.txt tester, meta tag previewer, status code checker, hash generator) and SSRF-guarded DNS/IP lookups.

Tool 2: [/confirm](#) — Social Proof, Trust & Local Engagement

Consolidates: 66socialproof, 66pusher, localboostai.

- **41 Social Proof Widget Types:** Conversion alerts, visitor counters, recent purchase notifications, video testimonials, emoji collectors, and score badges with geo/device targeting.
- **Web Push Marketing Engine (66pusher v23):** Subscriber segmenting, scheduled drip campaigns, recurring triggers, and automated RSS-to-push broadcasts.
- **LocalBoostAI Suite:**
 - *Review Booster:* Directs 4+ star reviews to Google/Facebook; routes negative reviews to private recovery forms.
 - *Coupons:* Local offer creation, claim collection, unique promo codes, and redemption tracking.
 - *Loyalty Cards:* Mobile QR stamp cards, visit tracking, reward milestones, and staff validation PINs.
 - *Lead & Feedback Forms:* In-page and modal widgets for quote requests, callbacks, waitlists, and private NPS surveys.
 - *Booking Pages:* Fast service booking pages with calendar availability, appointment slots, and automated reminders.

Tool 3: [/link](#) — Smart URLs, Dynamic QR, Bio-Links & File Transfers

Consolidates: 66biolinks, 66qrcode, 66transfer, linkqr, phpshort, droppey, swipple.

- **82+ Bio-Link Blocks:** Dynamic profile builder with embeds (Spotify, YouTube), product cards, Stripe/PayPal payment buttons, newsletter signups, and lead capture.
- **34+ Dynamic QR Code Configurations:** 25 body styles, 29 eye frame designs, embedded brand logos, dynamic color gradients, and vector export (SVG, EPS, PNG).
- **Enterprise URL Shortener:** Custom domains, deep-linking (iOS Universal Links, Android App Links), password protection, expiration dates, and A/B split rotation.
- **Branded Large File Transfers (Swipple + Droppey):** End-to-end AES-256 encrypted transfers, expiring links, custom backgrounds, and pluggable S3/Cloudflare R2 storage backends.

Tool 4: [/market](#) — Multimodal Ad Studio, Social Media & SEO Suite

Consolidates: magicads, magicai, open-seo-main, trends-checker-master, crawlseo-main, linkinator-main, stackposts.

- **All-in-One Ad Generation Studio (MagicAds):** Generates complete ad copy, headlines, hooks, and creatives tailored for 30+ ad platforms (Meta, Google, TikTok, LinkedIn, YouTube, X).
- **Multimodal Creative Engines:**
 - *AI Video Pro (Seedance 2.5):* High-converting vertical video ads, text-to-video, image-to-video, and avatar product demonstrations.
 - *AI Product Photoshoot (Nanobanana):* Professional studio product photography, scene compositing, background replacement, object removal, and style transfer.
 - *AI Voice & Dubbing (ElevenLabs):* Natural speech synthesis in 100+ languages, voice cloning, and audio dubbing.
- **Auto-Blogging & Content Wizard:** Generates 2,500-word SEO articles based on trending topics with AI humanization and auto-publishing to Shopify, WooCommerce, WordPress, and Ghost.
- **Unified Social Media Studio (Stackposts):** Visual calendar scheduling, bulk CSV uploads, watermarking, carousel creation, and AI comment/DM bots across 7 social platforms.
- **Semrush-Grade SEO & Competitive Radar:** DataForSEO keyword exploration, backlink analysis, competitor domain tracking, and Google Business Profile management.

Tool 5: [/monitor](#) — Uptime, API & Infrastructure Reliability

Consolidates: 66uptime, phpuptime.

- **Synthetic Multi-Location Monitoring:** Tests HTTP(S), TCP, UDP, ICMP (Ping), DNS, and SMTP across global probe nodes at intervals down to 10 seconds.
- **Real-Browser Synthetics:** Headless browser runs validating multi-step user checkout journeys with failure screenshot capture.
- **API Contract Validation:** Validates JSON schemas, status codes, and latency thresholds with assertion chaining.
- **Certificate & Domain Expiry:** Alerts at 30, 14, and 7 days before SSL/TLS certificate or WHOIS domain expiration.
- **Incident Management & Status Pages:** Branded public/private status pages with custom domains, maintenance windows, and multi-channel alert dispatch (Email, SMS, Slack, Discord, Teams).

Tool 6: [/track](#) — Privacy-First Analytics, Replays & Competitor Intelligence

Consolidates: 66analytics, phpanalytics, similarweb, stalkr, xtrabar.

- **Cookieless & Identified Telemetry:** GDPR/CCPA compliant tracking using cryptographically salted hashes that requires no cookie banners, migrating to identified user profiles upon login.
- **Session Replay Recording:** High-fidelity DOM event recording capturing cursor paths, clicks, scrolling, and inputs with client-side sensitive field masking.
- **Dynamic Heatmaps:** Click, mouse movement, and scroll-depth heatmaps across desktop, tablet, and mobile views.
- **Conversion Funnels:** Multi-step funnel visualization identifying exact drop-off stages.
- **Competitor Intelligence (SimilarWeb Grade):** Estimates traffic volume, referral sources, organic keywords, and demographic benchmarks for competitor domains.

4.1 Cross-Tool Ubiquitous Interoperability Bus

Tools communicate exclusively via [@mama1/bus](#) and standard resource identifiers:

```
// Universal Resource Name Structure:
// urn:mamal:<tool>:<workspace_id>:<entity_type>:<entity_id>
// Example: urn:mamal:audit:ws_01h8:site:site_abc123

export interface InteropEvent<T = unknown> {
  id: string; // UUIDv7
  workspaceId: string;
  sourceTool: 'audit' | 'confirm' | 'link' | 'market' | 'monitor' | 'track';
  eventType: string; // e.g. 'track.propensity.threshold_reached'
  resourceUrn: string;
  payload: T;
  timestamp: string; // ISO-8601 UTC
}
```

5. DESIGN SYSTEM, UI/UX & CUSTOMER EXPERIENCE (CX)

The design strictly implements the **Stripe-inspired aesthetic / indigo-ink ledger on frosted glass**:

- **Aesthetic Foundations:** Zero drop shadows. Elevation is conveyed through subtle background tint shifts (`#f8fafd` → `#ffffff`) and 1px hairline borders (`#e5edf5`). Strict 4px border-radius (`rounded-[4px]`).
- **Color Palette:** Indigo Ink (`#533afd` – primary CTA/active), Midnight Ink (`#061b31` – headings/metrics), Mist (`#f8fafd` – canvas), Frost (`#e5edf5` – borders), Periwinkle Wash (`#e8e9ff` – badges). High-contrast dark mode palette. WCAG 2.1 AA compliant (0 contrast violations).
- **Typography:** `sohne-var` / `Inter Tight`, weights 300 (light headlines) and 400 (body). Tracking tightens progressively as size grows (-1.40px at 56px). Enforces tabular numerals (`"tnum" on`) across all data tables and metrics.
- **Dual-Tier Navigation Shell:**
 - *Tier 1 (Rail — 64px width):* Workspace switcher, 6 master tool icons, Automations, Master AI switch, Account.
 - *Tier 2 (Contextual Drawer — 220px width):* Tool-specific navigation tree, resource lists, filter views, and quota counters.
 - *Global Top Bar:* ⌘K global search, workspace selector, AI engine health, quick "+ New" action dropdown.
- **Micro-Interactions & Motions:** Spring physics (stiffness 380, damping 30); toasts with 5-second countdown and immediate "Undo"; instant ⌘K command palette navigation; full `prefers-reduced-motion` compliance.
- **Mobile Responsiveness & PWA:** Fully responsive down to 320px width. Mobile transforms Tier 1 into an ergonomic bottom navigation bar and Tier 2 into a slide-over drawer. Installable Progressive Web App with offline service worker.

6. MULTI-MODEL AI ROUTING & REVENUE MARGIN ENGINE

All AI operations are orchestrated through a database-driven provider registry (`@mamal/ai`), enabling 1-to-3 click hot-swapping:

Specialized Capability	Default Configured Model	Hot-Swappable Alternatives
Behavioral Predictive Analytics	OpenAI Astra	Qwen Max, Claude 3.7 Sonnet
Closed-Loop Automation Engine	Anthropic Claude 3.7 Sonnet	GPT-5, Moonshot Kimi k1.5
Deep Reasoning & Attribution	DeepSeek Pro / R1	OpenAI o3, Google Gemini 2.5 Pro
Multimodal Video Generation	Seedance 2.5	Kling 3.0 Pro, Google Veo 3, Luma Dream Machine
High-End Product Photography	Nanobanana	Flux 2 Flex, Imagen 4, Stable Diffusion 3.5
Voiceover, Speech & Dubbing	ElevenLabs v3	Azure Cognitive Speech, Speechify
SEO, SERP & Backlink Data	DataForSEO API	Open-SEO Engine

AUTOMATIC 25% GROSS REVENUE MARGIN CALCULATION

Every AI request meters exact upstream API costs and automatically applies a mandatory 25% gross margin markup deducted in credits from the workspace wallet:

`User Credits Deducted = Math.ceil((Upstream Provider Cost USD × 1.25) × 100)`

Example: An upstream Seedance video generation costing \$0.10 is automatically billed to the user as \$0.125 (13 credits).

Master AI Kill-Switch: One global toggle immediately halts all external AI API calls workspace-wide. When disabled, all 6 tools continue running at 100% capacity using deterministic rule engines with zero credit consumption.

7. MONETIZATION, ENTITLEMENTS & PRICING ARCHITECTURE

- Free Tier (\$0/month — Zero Marginal Cost to Platform):** 18 client-side SEO tools, 1 basic uptime check, 5 short links, 1 social proof widget, and cookieless tracking up to 5,000 pageviews/month. Zero external AI API calls.
- Per-Tool Subscriptions (\$19 – \$49/month):** Dedicated access to a specific tool (e.g. `/audit` Pro or `/market` Pro) with generous resource quotas and a monthly baseline credit allowance.
- Unified Platform Subscription (\$99 – \$249/month):** Unrestricted access to all 6 tools, unlimited cross-tool automations, team collaboration, and high-volume monthly credit pools.
- Universal Credit Wallet (FIFO Buckets):** Credits operate as a universal platform currency. Plan credits expire monthly; purchased credit top-up packs rollover for 365 days.
- Lifetime Deal Tier (LTD — Strictly Non-Marginal):** One-time payment (\$299 – \$599) granting permanent access to all 6 core tools. *Strict Architectural Guard:* Excludes all free AI API usage. Lifetime users must connect their own API keys (BYOK) or purchase credit top-ups.

8. EXTENSION & PLUGIN ECOSYSTEM

Workspaces can toggle modular plugins from `/plugins` : **Affiliate System** (referral tracking & commission payouts), **Email Signature Generator** (HTML signatures with trackable links), **Image Optimizer** (automatic WebP/AVIF compression), **Newsletters Plugin** (segmented email broadcasts), **Push Notifications** (web push subscriber management), **PWA Generator** (instant mobile web app installation), and **Teams & RBAC** (granular role-based permissions).

9. RECOMMENDED TECH STACK & RUNTIME TOPOLOGY

Layer	Technology	Strategic Architecture Role
Monorepo & Build	Turborepo + pnpm	Cached incremental builds across tools and packages
Full-Stack App	Next.js 16 (React 19, RSC)	Unified dashboard, server actions, and public landing pages
Type Safety	TypeScript 5.8 + Zod 4	End-to-end strict contract validation at every API boundary
Relational Database	PostgreSQL 17 + Drizzle ORM	ACID transactional state for workspaces, billing, and entities
Event Telemetry Store	ClickHouse	Columnar store ingesting high-throughput clicks and analytics
Queues & Caching	Redis 7.4 (Valkey) + BullMQ	Distributed background workers (crawls, monitors, AI dispatch)
Edge Compute	Cloudflare Workers (Hono)	Sub-15ms short link redirects, tracking pixels, and QR resolution
Object Storage	Cloudflare R2	S3-compatible storage with zero egress fees for file transfers
Authentication	Better Auth	Multi-tenant organizations, OAuth, 2FA, and passkeys
Billing Engine	Stripe Billing + Paddle	Subscriptions, automated tax, and credit wallet checkout

10. SYSTEMATIC DEVELOPMENT QUALITY LOOPS

1. **Loop 1 — Design & Accessibility Gate (G1):** Verify all routes at 375px, 768px, 1280px, and 1920px. 0 horizontal overflows. 0 axe-core contrast/accessibility violations. All animations respect `prefers-reduced-motion`.
2. **Loop 2 — Ubiquitous UX & Workflow Gate (G2):** All actions and resources reachable via ⌘K palette in ≤3 keystrokes. Flawless keyboard focus management. Immediate undo toasts on destructive actions.
3. **Loop 3 — Cross-Tool Interoperability Gate (G3):** Bidirectional handover tests verified across all 6 tools via `@mamal/bus`. Isolated tool failure cannot degrade peer tools.
4. **Loop 4 — Operational Reliability Gate (G4):** Sub-15ms edge routing worldwide. SSRF security guards blocking all private IP ranges. Graceful AI rate-limit fallback with zero credit leakage.

LINEAR IMPLEMENTATION ROADMAP

- Step 1: Core Foundation (`@mamal/db`, `@mamal/bus`, `@mamal/ai`, `@mamal/ui`)
- Step 2: `/audit` Tool — 72-rule crawler + AI search visibility
- Step 3: `/confirm` Tool — 41 proof widgets + web push + LocalBoostAI
- Step 4: `/link` Tool — Bio-links + dynamic QR + branded transfers
- Step 5: `/market` Tool — MagicAds + Seedance 2.5 + Nanobanana + Stackposts
- Step 6: `/monitor` Tool — Multi-region probes + browser synthetics + status pages
- Step 7: `/track` Tool — ClickHouse telemetry + session replays + heatmaps
- Step 8: Predictive Engine & Visual Automation Canvas deployment